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ORIGINAL ARTICLE

Cybersecurity knowledge graphs construction and quality assessment

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Abstract
Cyber-attack activities are complex and ever-changing, posing severe challenges to cybersecurity personnel. Introducing knowledge graphs into the field of cybersecurity helps depict the intricate cybersecurity landscape and provides technical support for threat identification and situational awareness. However, during the process of constructing knowledge graphs, inevitable noise and conflicts may be introduced, leading to misleading inferences and decisions. This paper aims to address the issues of constructing and assessing the quality of cybersecurity knowledge graphs. We manually constructed a dataset of cybersecurity knowledge graphs (CS13K) and expanded the existing cybersecurity ontology, building a more versatile ontology applicable to existing data features. Building upon this, we utilized Neo4j to construct the cybersecurity knowledge graph. Furthermore, we propose an AttTucker model, based on Transformer, for assessing the quality of knowledge graphs. By employing numerous self-attention heads, this model captures latent information among entities and relations. While reducing the dimensionality of knowledge embeddings, the model achieves evaluation results comparable to high-dimensional embeddings. Finally, we incorporate the path-level information between entities in the knowledge graph into the evaluation of knowledge graph quality based on the AttTucker model. Experimental analysis on both generic datasets and cybersecurity datasets demonstrates that our model significantly improves F1 value and accuracy in knowledge graph quality assessment tasks, outperforming other models.

Keywords Cybersecurity · Knowledge graph · Quality assessment · Ontology construction · Attention mechanism

Introduction

In recent years, a vast amount of valuable cybersecurity information is scattered across the internet in a fragmented manner, such as vulnerability platforms, security information platforms, and security forums. However, these cybersecurity knowledge fragments have not been effectively integrated and utilized, limiting their unique advantages of big data knowledge. Additionally, with the rapid evolution of the cybersecurity landscape and the continuous innovation of attack methods, passive and static defense measures are no longer sufficient to meet the existing cybersecurity needs. Therefore, proactive analysis of cybersecurity-related information has become a necessary approach to address cybersecurity challenges. In light of the abundant and dispersed data and the diverse and highly complex malicious attacks confronted in the field of cybersecurity, knowledge graph technology is regarded as an effective solution. As an efficient knowledge management and reasoning tool, knowledge graphs can automatically discover potential associations between threat entities, provide valuable correlated information for cybersecurity situational awareness and risk assessment, associate intrusion alerts with attack paths, and propose optimal mitigation measures. Moreover, in the process of cyber-attack forensics, knowledge graphs can display

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vulnerable attack paths that require special attention. Analyzing this associative information, cybersecurity professionals can gain a better understanding of vulnerabilities and attack methods that adversaries may exploit, allowing them to strengthen defense measures and enhance overall cybersecurity protection. Therefore, constructing a comprehensive and accurate cybersecurity knowledge graph has become a necessary means to address cybersecurity challenges.

The construction of an ontology forms the foundation of building a cybersecurity knowledge graph. Acting as the knowledge system of the cybersecurity knowledge graph, cybersecurity ontology encapsulates the abstracted knowledge concepts of vulnerabilities, attackers, attack patterns, attack consequences, and defensive strategies in the security domain model. It can be utilized to represent, integrate, and share relevant cybersecurity knowledge, and can offer a formalized and unified terminology in the information security field along with the interrelationships among these terms. Simultaneously, it also has ontology reasoning and querying capabilities. Although several general-purpose ontologies in the cybersecurity field have been constructed to date, their complete applicability to a broader range of data sources is hampered due to differing focuses on cybersecurity elements. To address this issue, in this paper, we construct a new cybersecurity ontology, based on the analysis of CS13K data characteristics and the reuse of existing cybersecurity ontologies. This new ontology provides support for the construction of a comprehensive and accurate cybersecurity knowledge graph.

Knowledge graphs have played a crucial role in multiple domains, and an increasing number of software and applications are built upon them. However, when using knowledge graphs, it is often assumed that the content within these graphs is entirely error-free. If the content of the graph itself is flawed, it can significantly undermine the effectiveness of higher-level applications, leading to misleading reasoning and decision-making. However, noise and conflicts are inevitably introduced during the construction of knowledge graphs. With the widespread adoption of knowledge graphs, there is a growing demand for verifying the knowledge contained within them. In the early stages of knowledge graph development, the scale requirements were relatively low, allowing for manual inspection to determine the correctness of the triples in the graph. Nowadays, real-world applications rely on large-scale and extensive knowledge repositories, making manual fact-checking impractical. The quality assessment of knowledge graphs plays a vital role in enhancing their effectiveness and reliability in various application domains. Based on this, we propose a novel method for evaluating the quality of knowledge graphs, specifically focusing on assessing the quality of cybersecurity knowledge graphs.

In this paper, we manually constructed a cybersecurity knowledge graph dataset called CS13K, which exhibits high knowledge quality. Building upon a subset of existing cybersecurity ontologies, we developed a more versatile cybersecurity domain ontology that served as the foundation for constructing the cybersecurity knowledge graph. Inspired by the significant success of attention mechanisms in certain deep learning tasks [1, 2], we propose a Transformer-based model for evaluating the quality of knowledge graphs, named AttTucker. This model utilizes numerous self-attention heads to capture latent information between entities and relations. Furthermore, we incorporate the path-level information to assess the quality of the constructed cybersecurity knowledge graph. In summary, the key contributions of this paper are as follows:

(1). We manually constructed a dataset of cybersecurity knowledge graphs, named CS13K, which comprises 4494 entities, 12 relations, and 13,027 triples. CS13K represents the first fully manually constructed knowledge graph dataset in the field of cybersecurity.
(2). We expanded the UCO ontology from 8 categories to 16 categories and developed a new cybersecurity ontology suitable for the CS13K dataset. Leveraging this ontology, we constructed a cybersecurity knowledge graph that integrates diverse and heterogeneous cybersecurity data sources into semantically interpretable knowledge.
(3). We proposed a Transformer-based model for evaluating the quality of knowledge graphs. Under the condition of low-dimensional embeddings, this model achieves evaluation results comparable to high-dimensional embeddings, thereby addressing scalability issues in large-scale datasets.

The remaining sections of this paper are organized as follows: In Section "Related works", we discuss the related works. In Section "Methods", we introduce the cybersecurity knowledge graph dataset proposed in this study. We elaborate on the methodology used to construct the cybersecurity knowledge graph and provide an overview of our model. In Section "Results and discussion", we present the experimental results and provide a detailed evaluation and analysis of our model. Finally, we summarize the paper and discuss future research directions.

Related works

In this section, we reviewed the previous work in areas, such as cybersecurity datasets, ontology construction, knowledge graph construction, and knowledge graph quality assessment. We provided a comprehensive overview of research in these tasks within the field of cybersecurity.

Table 1 Information extraction and knowledge graph datasets in cybersecurity domain

Dataset	Purpose			Dataset size	Year	Remarks
	NER	RE	KG			
Ravendar [3]	√	–	–	46,300 tokens	2013	High-quality, manually annotated
Robert [4]	√	–	–	15,192 sentences	2014	Low-quality, automatically annotated
Kim [5]	√	–	–	498,000 tokens	2020	High-quality, large-scale
Wang [6]	√	–	–	175,220 tokens	2020	High-quality, manually annotated
Guo [7]	√	√	–	13,262 sentences	2021	High-quality, manually annotated
Rastogi [8]	–	–	√	40K triples	2021	Low-quality, automatically annotated
	–	–	√	3K triples	2021	High-quality, small-scale
Ours (CS13K)	–	√	√	13,027 triples	2022	High-quality, moderately large-scale

Cybersecurity datasets

Due to the high sensitivity and specialized nature of data in the field of cybersecurity, as well as the lack of large-scale specialized corpora, there has been limited research on cybersecurity knowledge graph datasets. As shown in Table 1, Robert et al. [4] constructed a cybersecurity entity recognition dataset using an automatic annotation method. They collected data from 15,192 cybersecurity sentences and used the "BIO" labeling scheme for data annotation. Kim et al. [5] manually annotated an entity recognition dataset in the cybersecurity domain, where domain experts labeled entities in CTI reports. This dataset contains a total of 498,000 words and was used for entity recognition research in the cybersecurity field using Bi-LSTM-CRF, achieving an F1 score of 75.05%. Wang et al. [6] manually annotated a dataset called DNRTI for named entity recognition research in the threat intelligence domain. The dataset was collected from over 300 threat intelligence reports, and entities from 13 categories were annotated. It contains a total of 175,220 words. Rastogi et al. [8] employed an automated extraction method to collect data from 1100 threat intelligence reports, constructing a malware knowledge graph dataset MT40K. This dataset contains 27,354 entities and 34 types of relations, comprising a total of 40,000 triples. As the extraction was automated, the accuracy of the triples in the dataset was not high. Based on the MT40K dataset, the authors constructed another dataset, MT3K, using manual filtering and expert review. The MT3K malware knowledge graph dataset contains 3000 triples, and while it boasts high triple accuracy, its smaller scale makes it less suitable for data-driven deep learning models. Guo et al. [7] manually constructed a cybersecurity information extraction dataset called OSINT by collecting information from various sources, such as vulnerability databases, security advisories, APT reports, security blogs, and hacker forums. They used this dataset for joint extraction of entities and relations in the cybersecurity domain, achieving an F1 score of 80.98%. However, due to extensive internal annotation work, the dataset has not been made publicly available. The use of natural language processing and deep learning methods for addressing cybersecurity issues is a relatively emerging research area. The lack of large-scale datasets in the cybersecurity field, coupled with the limited quality of existing publicly available datasets, has become one of the main obstacles restricting deep learning-based research in the cybersecurity domain.

Cybersecurity ontology

The construction of ontology serves as the basis for building a cybersecurity knowledge graph. Existing ontologies in the field of cybersecurity can be difficult to apply comprehensively to the integration of broader data sources due to their varying focuses during the construction process [9]. The earliest cybersecurity ontology was proposed by Undercoffer et al., with the aim of constructing an intrusion detection system for threat identification [10]. Since then, numerous ontologies have been proposed in the field of cybersecurity. Iannacone et al. [11] constructed a cybersecurity ontology encompassing 15 types of entities and 115 attributes. Later, Syed et al. [12] expanded this ontology, creating a more universal cybersecurity ontology, UCO, which can be mapped to the STIX format. The UCO ontology links a large number of cybersecurity standards and ontologies, such as CVE, CCE, CVSS, CAPEC, and more. It mainly includes eight types of entities, such as Attacker, Attack-Pattern, Consequences, and Indicator. In addition to mapping to STIX, UCO extends several related cybersecurity standards and ontologies, like CVE, CCE, CVSS, etc. Vorobiev et al. [13] built a Security Asset Vulnerability Ontology (SAVO) for modeling concepts, attributes, and interrelationships among network threats, risks, and vulnerabilities. Pingle et al. [14] enhanced the UCO cybersecurity ontology by adding crucial classes, such as Campaign, Tool, Course-of-Action, and also

expanded the relations, thereby completing the extension of the UCO ontology.

The construction of ontology can avoid the generation of redundant triples and accelerate the triple generation process in the knowledge graph. Therefore, building a comprehensive, appropriately granular, and semantically rich ontology is crucial for constructing a cybersecurity knowledge graph.

Knowledge graphs

In 2012, Google introduced the concept of a knowledge graph based on the Semantic Web. Essentially, a knowledge graph is a large-scale semantic network that encompasses various entities, concepts, and the semantic relations between entities. Currently, knowledge graphs have become a research focus in areas, such as semantic search, question answering systems, and recommendation systems [15, 16]. In this section, we first discuss the typical applications of knowledge graphs and then provide an overview of their applications in the field of cybersecurity.

Typical applications of knowledge graphs

From an application perspective, knowledge graphs can be categorized into general knowledge graphs and domain-specific knowledge graphs. General knowledge graphs focus on the breadth of knowledge and are widely used in intelligent search and recommendation scenarios. Typical examples of general knowledge repositories include Freebase, DBpedia, YAGO, and Wikidata. On the other hand, domain-specific knowledge graphs prioritize the depth of knowledge and serve specific query and analysis requirements within a particular knowledge domain. They have been extensively applied in domains, such as finance, healthcare, e-commerce, and academia.

Cybersecurity knowledge graphs

Cybersecurity issues are complex and intertwined, and multiple scholars have discussed security vulnerabilities [17, 18], attack patterns [19], and their corresponding solutions in various scenarios [20, 21]. Some scholars have also utilized federated learning techniques to address various cyber-attacks [22]. By introducing knowledge graphs into the field of cybersecurity, multiple heterogeneous and massive cybersecurity data sources can be integrated into semantically interpretable knowledge. Researchers can gain intuitive understanding of cybersecurity entities and their relations, such as the exploitation relation between hacker organizations and vulnerabilities, the affiliation relation between attackers and hacker organizations, and the inclusion relation between software and vulnerabilities. This integration can play a role in target profiling, APT detection, attack tracing, and provide new technical means for cybersecurity situational awareness, threat identification, and prediction.

Currently, research on knowledge graphs in the field of cybersecurity is still in its early stages, with limited studies on the overall technical framework for constructing and evaluating cybersecurity knowledge graphs. Yan et al. [23] utilized machine learning methods to extract entities in the field of cybersecurity and proposed a knowledge graph construction method applicable to the cybersecurity domain. Based on the constructed ontology, they completed the construction of a cybersecurity knowledge base. A cybersecurity situational analysis graph named Cygraph, launched by MITRE, constructs a knowledge graph from four dimensions: network infrastructure, security posture, cyber threats, and mission readiness. It is utilized to support tasks, such as attack surface identification and attack situational understanding, for the protection of critical assets [24]. Kiesling et al. [25] constructed a cybersecurity knowledge graph that integrates critical information from various public sources regarding vulnerabilities, weaknesses, and attack patterns. Their instance data is derived from designated standards and databases, such as CWE and CVE, by MITRE.

Knowledge graphs quality assessment

Knowledge graph quality assessment aims to quantitatively evaluate the quality of knowledge within a knowledge graph. Common methods for evaluating knowledge graph quality can be classified into two categories: detection methods using external data sources and detection methods based on internal data within the knowledge graph.

Detection methods using external data typically rely on external ontologies or network knowledge data to identify entity or relationship errors within the knowledge graph. For example, DeFacto et al. [26] used the DBpedia knowledge base as their research object, transformed the knowledge data into natural language statements, searched these statements in a browser, and determined the confidence values of the knowledge data based on the number of retrieved pages.

On the other hand, detection methods based on internal data of the knowledge graph typically model the knowledge graph as a graph and record paths between two entities through techniques such as random walks. If there are multiple paths between two entities in the knowledge graph, it suggests a high likelihood of a relations between them. Some scholars have also used knowledge representation learning methods to discover erroneous entities and relations. Konstantinos et al. [27] compared the effectiveness of graph embedding algorithms and path ranking algorithms and achieved good results by combining these algorithms in the task of knowledge graph error detection. Xie et al. [28] utilized the structural information within the knowledge graph for error detection. They introduced confidence scores

of triples into the knowledge representation learning process and proposed a confidence-aware knowledge representation learning method. Experimental results demonstrated that this method effectively captures the local and global structural information of the knowledge graph for triple confidence evaluation. Mendes et al. [29] proposed a knowledge quality assessment method based on the LDIF framework, allowing users to define quality assessment functions based on their specific business requirements or determine the final score through comprehensive consideration of multiple evaluation methods. Jia et al. [30] constructed a knowledge graph triple confidence evaluation model, assessing the confidence of triples from three dimensions: the entity level, the relation level, and the global level. They achieved good evaluation results on the FB15K dataset. Zhang et al. [31] employed automated machine learning technology (AutoML) to automatically design scoring functions (SFs) suitable for the triple classification problem in the knowledge graph. By establishing a unified SF representation and a search space, and using a greedy algorithm for efficient search, they achieved commendable results in the triple classification task.

Methods

Figure 1 presents the framework for constructing and evaluating the cybersecurity knowledge graph in this paper. It primarily involves four components: dataset construction, ontology construction, cybersecurity knowledge graph construction, and quality assessment of the cybersecurity knowledge graph.

Dataset

Cybersecurity threat reports present the technical and social aspects of cyber-attack events in the form of unstructured text. The construction of a cybersecurity knowledge graph first involves extracting key information from these reports, which is not a straightforward task. The existing open-source knowledge graph datasets in the field of cybersecurity are obtained through automatic extraction, and the accuracy of triples is not high, which is insufficient to reflect the actual situation of cybersecurity. The only known cybersecurity knowledge graph dataset with a higher accuracy rate, MT3K, is manually selected from MT40K. However, the scale of triples in this dataset is only 3000, which is not suitable for data-driven deep learning models. To truly reflect the status of cybersecurity, we have collected data from more than 440 public security reports and manually constructed a cybersecurity knowledge graph dataset. Each security report is issued by world-renowned security companies or government agencies and is analyzed, extracted, and annotated by several doctoral students from the Institute of Cyberspace Security who have good domain knowledge in cybersecurity. We named this dataset CS13K (Cybersecurity 13K). We will continue to update CS13K to expand its scale and scope, and reflect the true form of cybersecurity.

Table 2 Relation types in the CS13K dataset and their statistical characteristics

Relation	Size	Proportion (%)
BelongTo	989	7.59
Cause	924	7.09
Exploits	792	6.08
HasAttack Location	2068	15.88
HasAttackTime	688	5.28
HasCharacteristics	951	7.30
HasVulnerability	1010	7.75
Indicates	443	3.40
Mitigates	750	5.76
Targets	2259	17.34
Use	1489	11.43
Associate	664	5.10
Total	13,027	100.00

The CS13K dataset includes a total of 4,494 entities and 12 types of relations. These 12 types of relations are: *belongTo*, *cause*, *exploits*, *hasAttackLocation*, *hasAttackTime*, *hasCharacteristics*, *hasVulnerability*, *indicates*, *mitigates*, *targets*, *use*, and *associate*. The dataset contains a total of 13,027 triples, each with corresponding documents and sentences, which provides a data source for future work in areas, such as entity recognition and relation extraction. The statistical characteristics of the relation types in the dataset are shown in Table 2.

For example, the relation between ThreatActor and HackerGroup, Malware and MalwareFamily, malicious software and its type, and vulnerabilities and their type is denoted as “belongTo”. The relation between vulnerabilities and the aftermath of an attack is represented as “cause”. The relation between threat actors and vulnerabilities or weaknesses is represented as “exploits”. The relation between hacker groups and their attack locations, attack activities and their attack locations, and malicious software and their attack locations is represented as “hasAttackLocation”. The relation between hacker groups and their attack times, attack activities and their attack times, and malicious software and its attack time is represented as “hasAttackTime”. The relation between malicious software and its features or functions, hacker groups and their features or functions, and the features or functions of attack tools used by hacker groups is represented as “hasCharacteristics”. The relation between hardware, operating systems, and software and their own

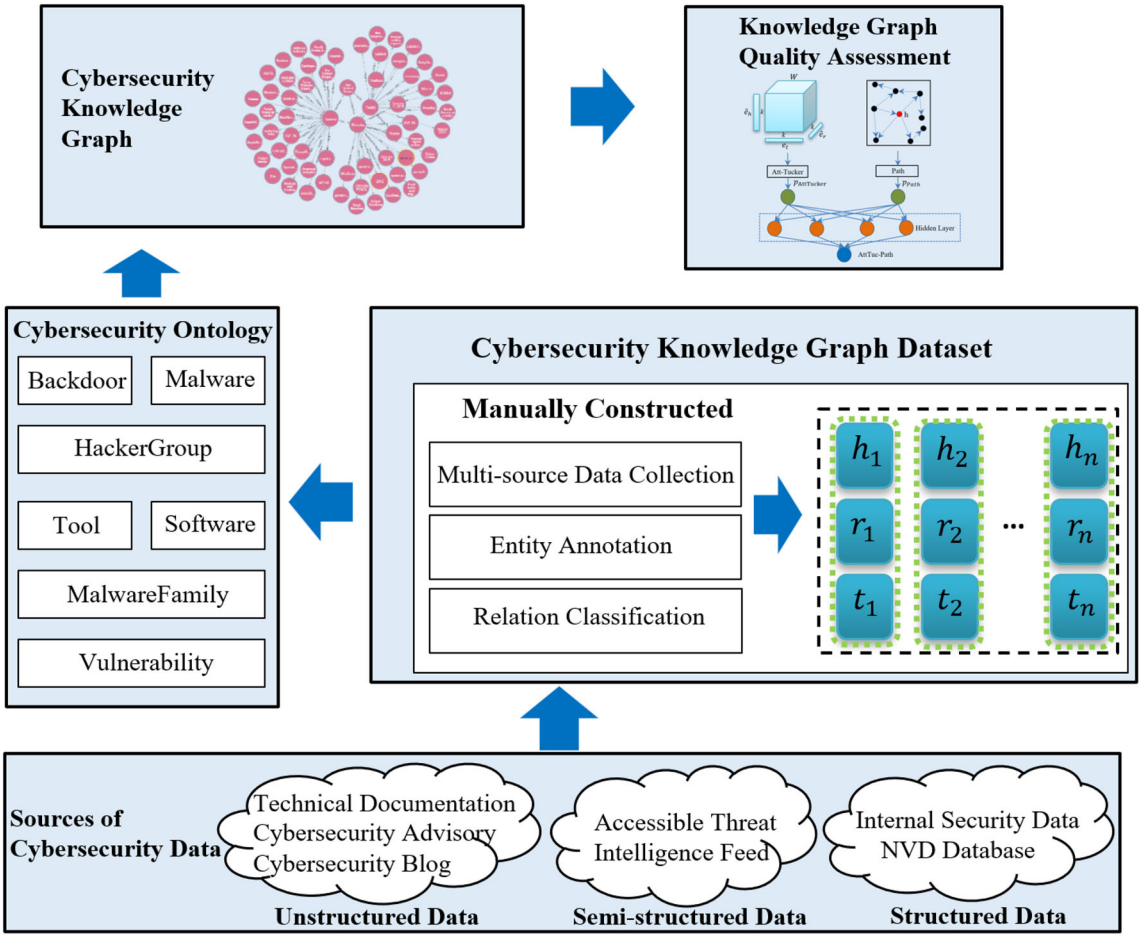

Fig. 1 Framework for cybersecurity knowledge graph construction and quality assessment

security vulnerabilities is represented as "hasVulnerability". The relation between hacker groups and their aliases is represented as "indicates". The relation between behaviors mitigating malicious software or vulnerabilities and the malicious software or vulnerabilities is represented as "mitigates". The relation between hacker groups, malicious software, or vulnerabilities and their targets is represented as "targets". The relation between attackers, attack activities, and the attack methods or tools they use are represented as "use". Furthermore, when two entities have a relation but do not fit the previous eleven categories, the label is set as "associate".

Construction of cybersecurity ontology

Based on the UCO ontology, we initially classify the knowledge related to cybersecurity to express the cybersecurity knowledge completely. We expand the classes related to cybersecurity attacks and preventive measures within the UCO ontology to 16, the descriptions of which are presented in Table 3. Subsequently, we define and constrain the relations between these classes. The relations and hierarchical structure among the classes constructed in this paper are illustrated in Fig. 2.

By integrating and instantiating the associations depicted in Fig. 2a–f, we obtained the schematic diagram of the network security ontology, as shown in Fig. 3. In Fig. 3, it can be observed that Hacker groups use Malware during attack activities, and the Malware may exploit various vulnerabilities present in operating systems or software, which can lead to a series of consequences. Typical consequences include denial of service, destruction of backups, memory corruption, and privilege escalation. The cybersecurity domain ontology serves as the foundation for constructing the knowledge graph, making it crucial to build a more comprehensive and versatile ontology suitable for existing data sources.

Cybersecurity knowledge graph

The knowledge graph can be defined as $KG = (h, r, t)$, where h and t denote the head and tail entity, respectively, and r represents the relation between h and t. In this study, the

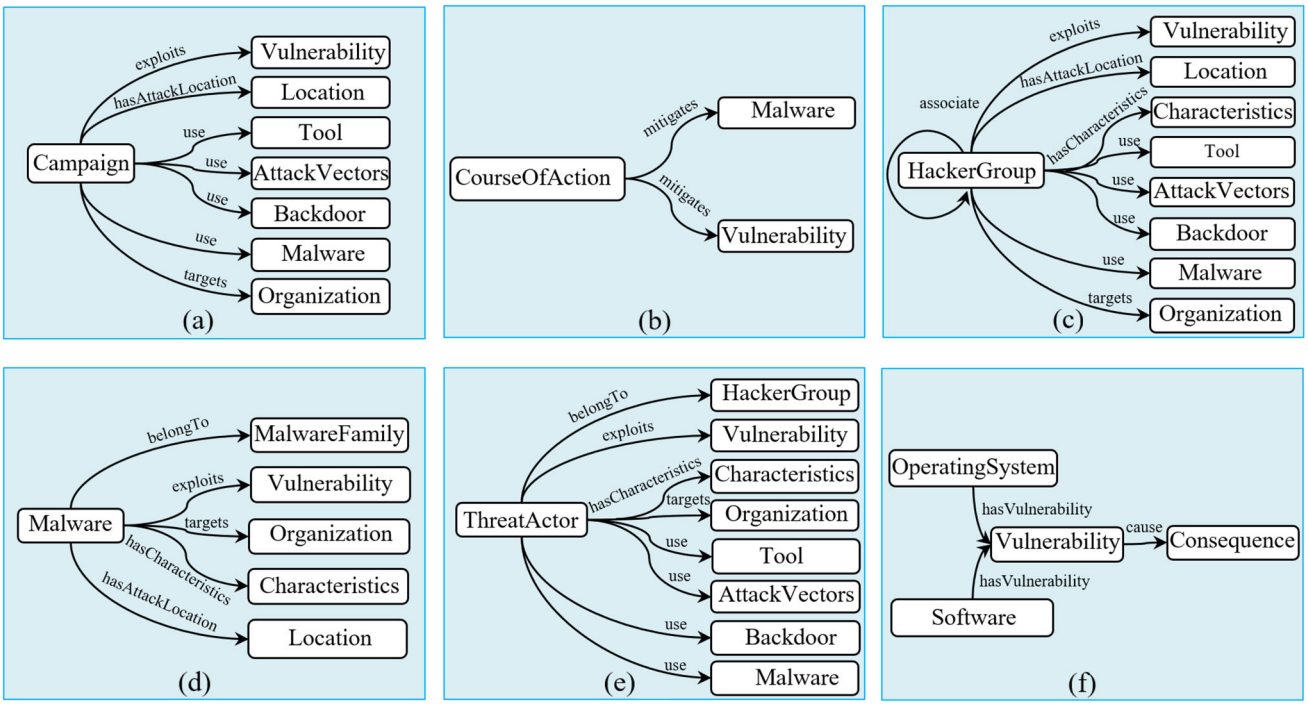

Fig. 2 Depiction of the schematic of relations among cybersecurity objects. Specifically, **a** represents the relations between *Campaign* and the other relevant classes; **b** presents the relations between *CourseOfAction* and the other related classes; **c** indicates the relations between *HackerGroup* and the other related classes; **d** expresses the relations between *Malware* and the other correlated classes; **e** displays the relations between *ThreatActor* and the other related classes; **f** shows the relations between *Vulnerability* and the other related classes

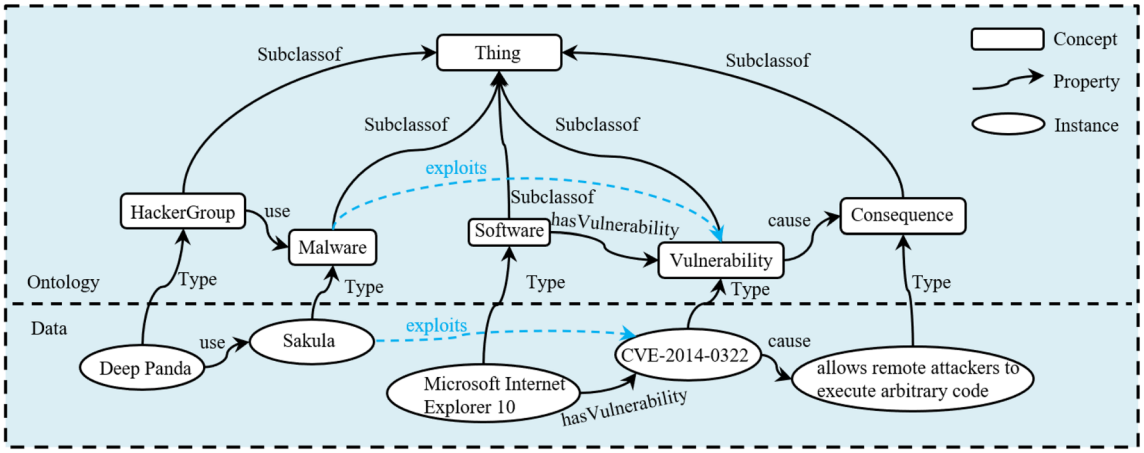

Fig. 3 Diagram of cybersecurity ontology

graph database Neo4j is selected as the software for storing cybersecurity knowledge data. Neo4j is currently the most widely used and popular graph database software, organizing and storing data in the form of nodes and edges. By incorporating vulnerabilities, attacks, defense mechanisms, etc., into the knowledge graph, it becomes possible to associate intrusion alerts with attack paths, visually presenting attack methods, indicators, and consequences in the form of a graph, and proposing the best mitigation measures. During the process of investigating cyber-attacks, it is possible to identify the paths that are vulnerable and require focused attention and create an overall picture of the attack activities, facilitating analysis, detection, classification, and attribution of various cyber threats. Figure 4 is a partial diagram of the cybersecurity knowledge graph built using Neo4j.

The knowledge graph offers a fresh perspective for understanding and addressing cybersecurity threats. For example,

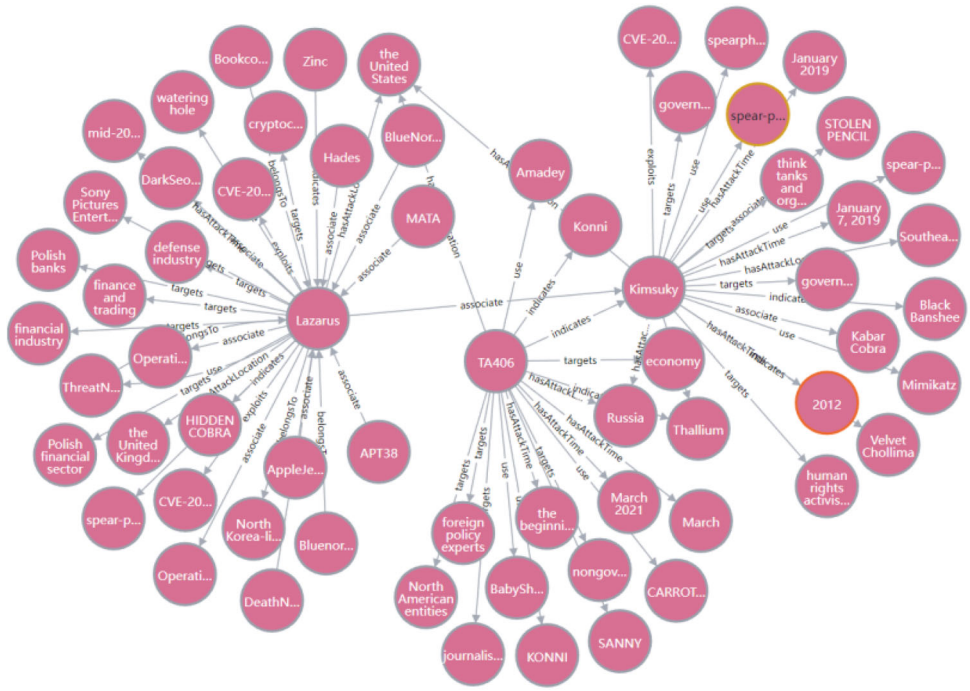

Fig. 4 Diagram of cybersecurity knowledge graph

in the case of a common malware cyber-attack, the knowledge graph can classify it into multiple subclasses (such as Trojan, worm, spyware) and link it to specific attack strategies (such as ransomware, user data theft). Furthermore, it can be associated with specific systems and applications at risk (such as operating systems, web servers) and relevant defense measures (such as firewall configurations, encryption, and intrusion detection systems). Analyzing the information in the knowledge graph, cybersecurity experts can make more accurate predictions and prevent various cyber-attacks.

Moreover, taking automated threat detection and response as an example, the knowledge graph contains extensive information about past cyber-attacks, such as attack types, methods, affected systems, attack timestamps, and impact of the attacks. Based on this information and the success achieved by intelligent computing [32], optimization algorithms [33, 34], and machine learning algorithms in various scenarios and tasks [35, 36], we can train a machine learning model based on this information. When new cyber activities (such as log entries, cyber traffic) are inputted into the model, it can identify threats that match the attack patterns in the knowledge graph. This enables us to provide early warning and response before the attacks pose a real threat to the system.

Quality assessment of cybersecurity knowledge graphs

To the best of our knowledge, no model has attempted to address the problem of quality assessment of knowledge graphs under the condition of low-dimensional embedding. Existing knowledge representation models have achieved good results in many downstream tasks using high-dimensional embeddings, whereas the performance with low-dimensional embeddings needs to be improved. This has led to the scalability issue of existing models in large-scale datasets. Therefore, we propose a Transformer-based knowledge graph quality assessment model (AttTucker) that utilizes a large number of self-attention heads to capture latent information between entities and relations. Furthermore, we integrate path-level information into the AttTucker model to evaluate the confidence of triples in the cybersecurity knowledge graph.

Based on the earliest research on attention mechanisms [1], during the encoding phase, for each triple (h, r, t), we set the dimension of the head entity as d_h, the dimension of the tail entity as d_t, and the dimension of the relation as d_r. To ensure the validity of tensor calculations in the model, we consider setting $d_h = d_t = d_r = k$. We define the embeddings of the head entity h and the relation r as $e_h \in R^k$ and $e_r \in R^k$, respectively. The concatenated vector $(e_h; e_r)$ is then constructed as the query $Q \in R^{2k}$, value $V \in R^{2k}$, and key $K \in R^{2k}$. Based on the head entity h and the relation r, we can obtain the output matrix $H \in R^{2k}$ as follows:

$$H = A_{hr} V = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V, \tag{1}$$

Table 3 Classes related to cybersecurity and their descriptions

Class	Description
AttackVectors	A means or method through which hackers can gain access to a computer or network server
Backdoor	A method of logging in to the system, which can bypass the existing security settings of the system
Campaign	A series of malicious activities targeted at a specific objective or launched within a certain period of time
Characteristics	Characteristics of hacker organizations, attack activities, malware, etc
Consequence	Consequences resulting from the exploitation of vulnerabilities
CourseOfAction	Measures employed to prevent or respond to attacks
HackerGroup	A group or organization engaged in malicious operations
Location	The location where the cyber-attack takes place
Malware	Utilized to compromise the confidentiality, integrity, or availability of the victim's data or systems
MalwareFamily	Family of Malware
OperatingSystem	Operating system
Organization	The target of an attack, such as an organization or company
Software	Software
ThreatActor	Individuals who conduct malicious operations
Tool	Tools utilized by HackerGroup, Campaign, ThreatActor, etc., for carrying out attacks
Vulnerability	A bug in the software that can be directly exploited by hackers to gain access to a system or network

where $A_{hr} \in R^{2k \times 2k}$ are the attention matrices. When using multiple attention heads, the multi-head attention can be represented as:

$$\text{Multi}H(Q, K, V) = \left[H_1, \ldots, H_j\right] W_O, \tag{2}$$

where $H_i = \text{softmax}(\frac{(QW_i{}^Q)(KW_i{}^K)}{\sqrt{d_k}})(VW_i{}^V)$. $W_i{}^Q \in R^{k \times d_k}$, $W_i{}^K \in R^{k \times d_k}$, $W_i{}^V \in R^{k \times d_v}$, and $W_O \in R^{hd_v \times k}$ are projection matrices, and h is the number of heads.

Furthermore, we use a token-wise feed-forward to generate the final output representation H_{output} for each triple of the knowledge graphs:

$$H_{\text{output}} = FFN(\text{Multi}H(Q, K, V)), \tag{3}$$

where $FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2$, and $W_1 \in R^{k \times d_h}$, $b_1 \in R^{d_h}$, $W_2 \in R^{d_h \times k}$, $b_2 \in R^k$ are learnable parameters.

During the decoding phase, based on the simplicity and efficiency of the Tucker model [37], we construct the knowledge graph quality assessment model on the basis of the Tucker model. Let $H_{\text{output}} = (\widetilde{e}_h; \widetilde{e}_r)$, where $\widetilde{e}_h$ and $\widetilde{e}_r$ are the output representations of the head h and the relation r. The scoring function for the triple (h, r, t) can be represented as:

$$\varphi(h, r, t) = W \times_1 \widetilde{e}_h \times_2 \widetilde{e}_r \times_3 e_t, \tag{4}$$

where $W \in R^{k \times k \times k}$ is the core tensor, for each score $\varphi(h, r, t)$, We apply the logistic sigmoid function to each score $\varphi(h, r, t)$ to obtain the predicted probability p of a triple being true:

$$p = \text{sigmod}(\varphi(h, r, t) - th_r), \tag{5}$$

where th_r is the threshold that partitions triples into positive and negative examples, which is a relation-specific parameter. In the CS13K dataset, we have 12 different th_r values. When $p \geq 0.5$, the triple is judged as positive, and when $p < 0.5$, the triple is classified as negative.

We use Binary Cross-Entropy (BCE) loss to train the model, minimizing the following loss function.

$$L = -\frac{1}{V} \sum_{i \in V} (y_i \log p_i + (1 - y_i) \log(1 - p_i)), \tag{6}$$

where V is the set of all entities, p_i is the predicted probabilities of triples, and y_i is the binary label.

In addition, based on the work in the literature [30], we incorporate the path-level information into the AttTucker model (referred to as AttTuc-Path model). As shown in Fig. 5, we denote the triple confidence obtained from the AttTucker model as $p_{\text{AttTucker}}$, and the triple confidence obtained using path-level information as p_{Path}. We concatenate $p_{\text{AttTucker}}$ and p_{Path} to form a vector V, which is then fed into a multi-layer perceptron to obtain the final probability value p indicating the confidence of a triple being true.

$$V = [p_{\text{AttTucker}}; p_{\text{Path}}], \tag{7}$$

$$h_n = \sigma\left(W_{h_n} V + b_{h_n}\right), \tag{8}$$

$$p = \alpha(W_o h_n + b_o), \tag{9}$$

Here, h_n represents the n-th hidden layer, W_{h_n} and b_{h_n} denote the learnable parameters in the n-th hidden layer. W_o and b_o are the learnable parameters of the output layer.

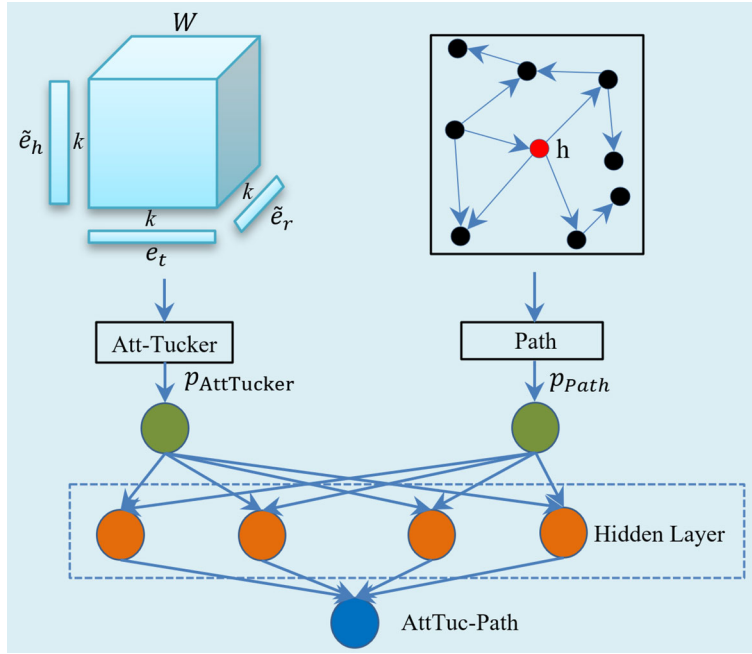

Fig. 5 The AttTuc-Path model for knowledge graph quality assessment

Table 4 The parameter numbers and time complexity

Models	Numbers of parameters	Time complexity
TransE	$Ek + Rk$	$O(k)$
TransH	$Ek + 2Rk$	$O(k)$
TransD	$2Ek + 2Rk$	$O(k)$
TransR	$Ek + Rk(k + 1)$	$O(k^2)$
Tucker	$Ek + Rk + k^3$	$O(k^3)$
AttTucker	$Ek + Rk + 3k^2 + k^3$	$O(k^3)$

Time complexity and parameter numbers analysis

We conducted an analysis of the model complexity and parameter numbers. Table 4 presents the parameter numbers and time complexity of the Tucker and AttTucker models developed in this study, as well as the baseline models TransE, TransH, TransD, and TransR. In the table, E represents the number of entities in the knowledge graph, R represents the number of relation types, and k represents the embedding dimension. From Table 4, it can be observed that the parameter numbers of the baseline models exhibits a certain correlation with the number of entities, the number of relations, or the embedding dimension.

In the task of knowledge graph quality assessment, we represent the triples in the knowledge graph in the form of a third-order Tucker decomposition:

$$X = G \times_1 A \times_2 B \times_3 C, \tag{10}$$

Here, X is a third-order tensor, G is a core tensor, and A, B, and C are factor matrices. A and C have dimensions (E, k), B has dimensions (R, k), and G has dimensions (k, k, k). This means that A and C represent entity embeddings, B represents relation embeddings, and G captures the interaction between these embeddings. Therefore, the parameter count of the Tucker decomposition is:

$$Params_{Tucker} = Ek + Rk + k^3. \tag{11}$$

The main cost here is the core tensor G, which has a size of k^3, and can become computationally expensive when k is large. The costs of A and C are proportional to the number of entities E, and the cost of B is proportional to the number of relations R.

In the multi-head self-attention mechanism, for each attention head, we have three weight matrices: query (Q), key (K), and value (V). Assuming we have h heads, a model embedding dimension of k, and each head has a dimension of d_k, then for each attention head, the sizes of the Q, K, and V matrices are all (d_k, k).

Therefore, the parameter numbers of a single self-attention head is $3 * d_k * k$, and since there are h heads, the total parameter count for self-attention is:

$$Params_{attention} = 3 * d_k * k * h \tag{12}$$

$$= 3k^2.$$

Therefore, the total parameter numbers of the AttTucker model, including both the Tucker decomposition and multi-head self-attention parameters, can be represented as:

$$\text{Params}_{\text{AttTucker}} = \text{Params}_{\text{Tucker}} + \text{Params}_{\text{attention}} \tag{13}$$

$$= Ek + Rk + 3k^2 + k^3.$$

As a result, the time complexity of the Tucker and AttTucker models developed in this study is $O(k^3)$. The time complexity of TransE, TransH, and TransD is $O(k)$, and the time complexity of TransR is $O(k^2)$. It can be observed that, under the assumption of fixed embedding dimensions for entities and relations, our model contains more parameters. However, in the case of low-dimensional embeddings, our model can achieve evaluation results comparable to those with high-dimensional embeddings. Furthermore, when dealing with large-scale datasets, the parameter count introduced by the embedding dimension is negligible compared to the number of entities and relations. Therefore, our model addresses the scalability issue to some extent in large-scale datasets.

Results and discussion

In this section, we evaluate the quality assessment performance of the Tucker, AttTucker, and AttTuc-Path models on the general-domain knowledge graph dataset FB15K. We also verify the performance of our model on the cybersecurity dataset CS13K.

Experimental settings and evaluating metrics

The FB15K dataset is a widely used general-domain knowledge graph dataset, consisting of 14,951 entities and 1345 relations. Since the FB15K dataset does not contain negative triples, we adopt the method proposed in reference [38] to construct negative triples by replacing either the head or tail entity, simulating the construction of real-world knowledge graphs. To investigate the impact of noise ratios in the knowledge graph on model performance, we construct datasets with noise ratios of 10%, 20%, 30%, 40%, and 50%, denoted as FB15K-N10, FB15K-N20, FB15K-N30, FB15K-N40, and FB15K-N50, respectively. Similarly, for the cybersecurity dataset CS13K, we also construct datasets with noise ratios of 10%, 20%, 30%, 40%, and 50%, denoted as CS13K-N10, CS13K-N20, CS13K-N30, CS13K-N40, and CS13K-N50, respectively. For all datasets, we divide them into training and testing sets in a 4:1 ratio. Accuracy and F1 score are used as evaluation metrics. For a given triple, it is classified as a positive sample if the predicted probability $p \geq 0.5$, otherwise it is classified as a negative sample. During the prediction process, there are four possible scenarios:

1. When the actual label is negative and the prediction is also negative, it is a true negative (TN).
2. When the actual label is negative but the prediction is positive, it is a false positive (FP).
3. When the actual label is positive but the prediction is negative, it is a false negative (FN).
4. When the actual label is positive and the prediction is also positive, it is a true positive (TP).

Therefore,

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{14}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{15}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{16}$$

$$F1 = \frac{2\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{17}$$

We use Accuracy and F1 score as evaluation metrics, and analyze the Precision and Recall values to better reflect the model's ability to distinguish between positive and negative samples. The main parameter settings for the AttTucker model are shown in Table 5.

Table 5 Main parameter settings for the AttTucker model

Batch size	2000
Learning rate	0.015
Epochs	3000
Number of heads	128
Encoder dropout	0.7
Position feed-forward dropout	0.7
Multi-heads attention dropout	0.1
Dimension embeddings	100

Experimental results and evaluation

Performance on general knowledge graph dataset

On the FB15K dataset, we compared our model with other baseline models, including NTN, CKRL, TransT, DistMult, Analogy, ComplEx, SimplE, TransE, TransR, and others. The results are shown in Table 6. It can be observed that both the Tucker and AttTucker models outperform all the baseline models in terms of accuracy and F1 value. Notably,

Table 6 Results of different models on FB15K

Models	Accuracy	F1 value
NTN [38]	0.665	–
CKRL(LT) [28]	0.818	–
CKRL(LT + PP) [28]	0.819	–
CKRL(LT + PP + AP) [28]	0.817	–
TransT (TT) [39]	0.822	–
TransT (TT + DT) [39]	0.824	–
DistMult [31]	0.808	–
Analogy [31]	0.821	–
ComplEx [31]	0.818	–
SimplE [31]	0.815	–
AutoSF [31]	0.827	–
TransE [30]	0.868	0.876
TransH [30]	0.912	0.913
TransD [30]	0.913	0.913
TransR [30]	0.902	0.904
Tucker	0.945	0.948
AttTucker	0.955	0.956

the TransE model does not achieve high evaluation results due to its limitations in handling 1-to-N, N-to-1, and N-to-N relations. By introducing more complex relation-specific parameters, models, such as TransH, TransR, and TransD, have achieved better evaluation results than the TransE model. The AttTucker model achieves an F1 value of 0.956 on the FB15K dataset, which is 5.2% higher than the F1 value of the TransR model. This indicates that modeling the knowledge graph as a core tensor and three factor matrices, along with the introduction of multi-head attention mechanism, enables the model to better capture the complex relations in the knowledge graph and improve its accuracy.

Furthermore, we analyzed the impact of the proportion of noisy triples in the knowledge graph on the model's performance, and the results are shown in Table 7. From Table 7, it can be observed that as the proportion of noisy triples in the knowledge graph increases from 10 to 50%, both the Tucker and AttTucker models show a decrease in evaluation performance. The reason for this could be that as the proportion of noise increases, the triples in the knowledge graph may become more inconsistent. For example, there may be noisy triples that contradict other correct triples. This inconsistency can confuse the model during inference and prediction, leading to a decrease in its performance. It is evident from Table 7 that although the model's performance decreases with the increase in the proportion of noise, the evaluation results remain at a good level. This indicates that while the presence of noise does have some impact on the model's performance, our model is still able to maintain relatively stable performance, demonstrating its robustness to noisy triples.

Table 7 Impact of noise ratio on model performance in the FB15K dataset

Models	Datasets	Accuracy	F1 value
Tucker	FB15K-N10	0.971	0.973
	FB15K-N20	0.968	0.970
	FB15K-N30	0.962	0.964
	FB15K-N40	0.954	0.955
	FB15K-N50	0.945	0.948
AttTucker	FB15K-N10	0.979	0.980
	FB15K-N20	0.974	0.975
	FB15K-N30	0.969	0.971
	FB15K-N40	0.961	0.963
	FB15K-N50	0.955	0.956

To validate the effectiveness of the AttTucker model in the knowledge graph quality assessment task, we conducted a visual analysis of the evaluation results of the AttTucker model on the FB15K-N50 testing sets, as shown in Fig. 6. Similar results can also be obtained when using the Tucker model. Figure 6a illustrates the confidence evaluation results for negative triples, while Fig. 6b shows the confidence evaluation results for positive triples. By observing Fig. 6a, we can see that negative triples generally have low confidence values, and these values are mostly below 0.5. Conversely, Fig. 6b demonstrates that positive triples generally have high confidence values, and these values are mostly above 0.5. Such visual analysis helps us intuitively understand the model's predictive ability for triple confidence in the knowledge graph, further confirming the superiority and reliability of the AttTucker model in the task of knowledge graph quality assessment.

Performance on the cybersecurity dataset

In addition, we evaluated the performance of Tucker and AttTucker models on the cybersecurity knowledge graph dataset CS13K. The results are shown in Table 8. From Table 8, it can be observed that for the TransE model, as the noise ratio increases, the accuracy and F1 value gradually decrease. The noise ratio affects the performance of the TransE model on the cybersecurity dataset, resulting in lower accuracy and F1 value. The Tucker model demonstrates higher accuracy and F1 value on the CS13K dataset at different noise ratios. In contrast, the AttTucker model exhibits better performance on the CS13K dataset at different noise ratios. Although the accuracy and F1 score of the AttTucker model decrease with

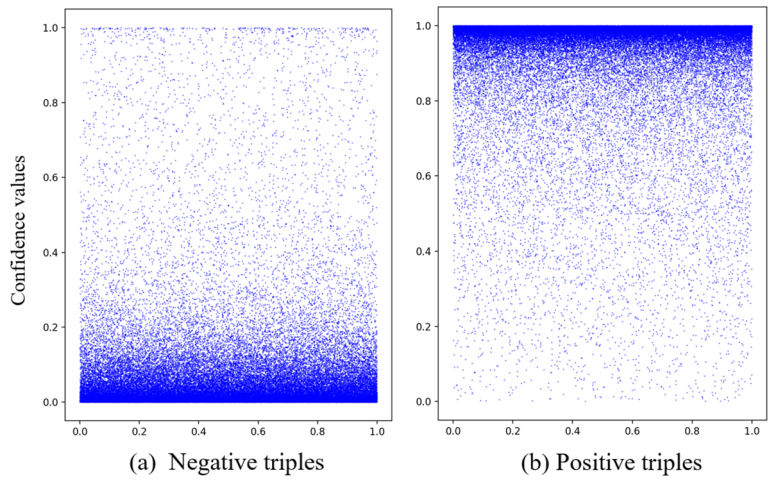

Fig. 6 Confidence distribution of positive and negative triples of FB15K-N50

Table 8 Impact of noise ratio on model performance in the CS13K dataset

Models	Datasets	Accuracy	F1 value
TransE	CS13K-N10	0.881	0.882
	CS13K-N20	0.870	0.872
	CS13K-N30	0.864	0.866
	CS13K-N40	0.859	0.860
	CS13K-N50	0.854	0.855
Tucker	CS13K-N10	0.951	0.952
	CS13K-N20	0.935	0.938
	CS13K-N30	0.922	0.923
	CS13K-N40	0.911	0.913
	CS13K-N50	0.893	0.894
AttTucker	CS13K-N10	0.965	0.968
	CS13K-N20	0.950	0.951
	CS13K-N30	0.938	0.940
	CS13K-N40	0.925	0.926
	CS13K-N50	0.908	0.909

Table 9 The quality evaluation results for different relation types

Relations	Precision	Recall	F1 value
BelongTo	0.868	0.883	0.875
Cause	0.916	0.931	0.923
Exploits	0.882	0.836	0.858
HasAttackLocation	0.893	0.898	0.895
HasAttackTime	0.953	0.946	0.949
HasCharacteristics	0.913	0.906	0.909
HasVulnerability	0.911	0.906	0.908
Indicates	0.902	0.913	0.907
Mitigates	0.893	0.907	0.900
Targets	0.956	0.948	0.952
Use	0.917	0.899	0.908
Associate	0.909	0.926	0.917
Average	0.909	0.908	0.909

increasing noise ratio, it maintains stronger robustness compared to the TransE model, enabling it to maintain higher performance in noisy environments.

We also reported the precision, recall, and F1 value of the AttTucker model for each specific relation type. We used the CS13K-N50 dataset, which has a higher noise ratio, for the quality evaluation. The results are shown in Table 9. By analyzing the precision, recall, and F1 value for each relation type, we can gain deeper insights into the model's quality assessment performance across different relation types. We found that the model exhibits varying precision and recall across different relation types. For example, relation types, such as "hasAttackTime" and "targets", have relatively higher precision and recall, while relation types like "belongsTo" and "exploits" have relatively lower precision and recall. This indicates that the model's quality assessment capability varies for different relation types, possibly due to variations in sample features among different relations in the dataset.

Similarly, we visualized the confidence evaluation results of the AttTucker model on the CS13K-N50 testing sets, as shown in Fig. 7. Figure 7a illustrates the confidence evaluation results for negative triples, while Fig. 7b shows the confidence evaluation results for positive triples. From Fig. 7, we observed similar results as in the general dataset: the confidence values for negative triples are generally low, with most values below 0.5. In contrast, the confidence values for positive triples are generally high, with most values above 0.5. This indicates that the model performs well in quality assessment of cybersecurity knowledge graphs.

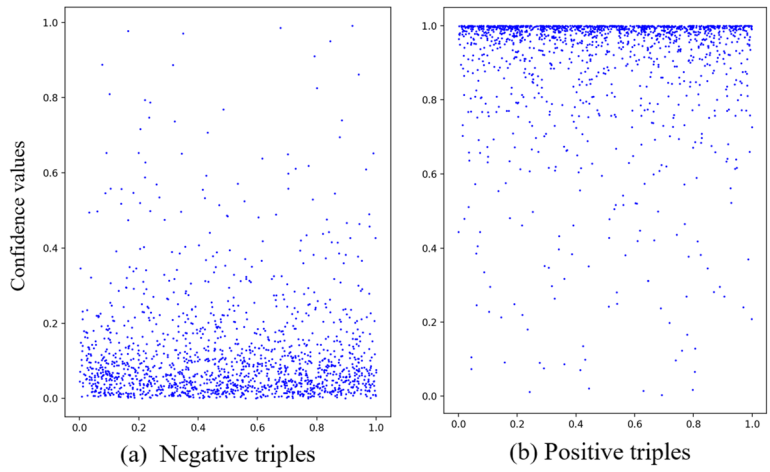

Fig. 7 Confidence distribution of positive and negative triples of CS13K-N50

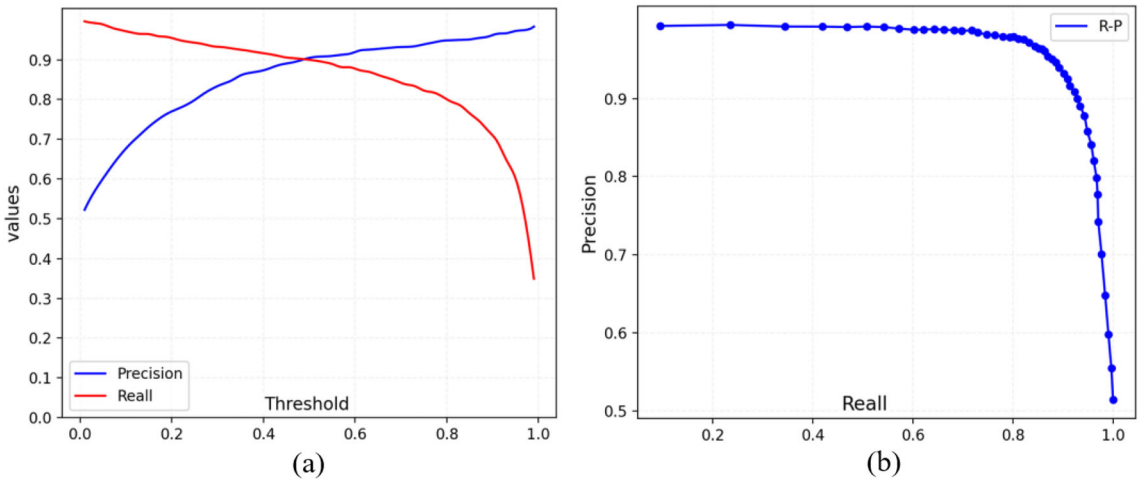

Fig. 8 **a** Precision and recall values for different thresholds. **b** The Recall–Precision curves

Moreover, we utilized precision and recall curves to represent the model's evaluation performance of triple confidence scores. The threshold used for triple confidence discrimination is adjustable, where triples with confidence scores higher than the threshold are classified as positive, and those below it are classified as negative. As depicted in Fig. 8a, as the threshold increases, precision continually improves while recall decreases. At lower thresholds, recall is higher, but as the threshold increases, recall decreases. When the threshold surpasses 0.5, recall starts to decline significantly, and the closer the threshold is to 1, the more rapid the decline. This indicates that the confidence scores of positive samples in the CS13K-N50 dataset are generally above 0.5, and when the threshold is higher than 0.5, precision remains at a high level, indicating that the model has good capability to identify negative samples in the CS13K dataset. Further analysis of this phenomenon suggests that the higher precision and lower recall results may stem from stricter discrimination requirements for negative samples. With higher thresholds, the model tends to classify triples with lower confidence scores as negative, thereby improving precision. However, this also leads to some positive triples being mistakenly classified as negative, resulting in lower recall.

On the other hand, the Recall–Precision curve in Fig. 8b shows that when the recall is less than 0.8, as the recall increases, precision remains almost constant and maintains a high value. This may be due to the model's cautious selection of positive triples at lower recall rates to ensure higher precision. However, as the recall increases, the model begins to classify more triples as positive, including some triples with lower confidence scores, leading to a rapid decrease in precision. The balancing point of the curve is approximately around a recall of 0.90, indicating that at this point, the model achieves high precision while maintaining a good recall rate.

The use of attention mechanisms to capture interactions and dependencies between entities and relations in the knowledge graph is a key aspect of the AttTucker model. Therefore, we investigated the impact of multi-head attention on the model's performance. As shown in Fig. 9, within a certain range, increasing the number of attention heads enhances the

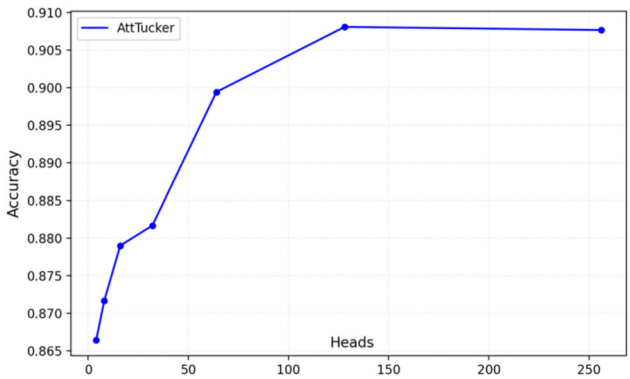

Fig. 9 The impact of the number of heads on the model's accuracy

accuracy of the model for knowledge graph quality assessment. However, we also observed that as the number of heads continues to increase, the performance improvement of the model gradually diminishes until reaching a saturation state. This phenomenon may be attributed to the working principle of the multi-head attention mechanism. A smaller number of heads may not fully capture the complex interaction relationships in the knowledge graph, limiting the expressive power of the model. Increasing the number of attention heads can enhance the model's modeling capability for different entities and relations, thus improving the accuracy. However, based on the results in Fig. 9, we observed that after the number of heads reaches 128, further increasing the number of heads does not lead to significant performance improvements. This suggests that the effective range of the multi-head attention mechanism has been reached, and additional heads do not provide additional information gain. These findings provide guidance for further optimization and design of attention mechanisms to achieve better knowledge graph quality assessment performance.

Additionally, we also verified the effectiveness of the model under low-dimensional embeddings to demonstrate its scalability when evaluating large-scale knowledge graphs. Table 10 displays the evaluation results of the Tucker model and the AttTucker model on the CS13K-N50 dataset at different embedding dimensions. It can be observed that as the embedding dimension increases, the evaluation accuracy of the model gradually improves. When the dimension is below 100, increasing the embedding dimension further enhances the model's accuracy. Moreover, the comparative results also show that, at the same embedding dimension, the AttTucker model outperforms the Tucker model in terms of evaluation accuracy. This indicates that the attention mechanism can indeed better capture the interaction information between entities and relations, thereby improving the evaluation accuracy of the model under low-dimensional embeddings. It is worth noting that at the dimension of 75, the AttTucker model achieves a accuracy level comparable to that of the Tucker model at the dimension of 100. This suggests that the addition of the attention mechanism can compensate for some information loss under low dimensions and significantly enhance the model's performance. Furthermore, we observed that at the dimension of 64, the evaluation accuracy of the AttTucker model surpasses that of the Tucker model at the dimension of 75. This further validates the effectiveness of the attention mechanism in improving the model's performance under low dimensions.

Besides, we also tested the impact of path-level information between entities in the knowledge graph on the performance of the knowledge graph quality assessment model. In this experiment, we again selected the CS13K-N50 dataset with a higher noise ratio for evaluation and used the accuracy as the evaluation metric. We specifically focused on examining the performance of path-level information in knowledge graph quality assessment. The evaluation results are presented in Table 11, where "Path" represents the evaluation accuracy considering only the path-level information from the CS13K-N50 dataset, and "AttTuc-Path" represents the final knowledge graph quality assessment accuracy obtained by integrating the AttTucker model with the path-level information.

From Table 11, it can be observed that combining path-level information with the AttTucker model significantly improves the model's predictive performance. By incorporating path-level information, the model is able to comprehensively consider the semantic associations between entities within the knowledge graph, thereby enhancing the model's predictive and assessment capabilities for triple confidence scores in the knowledge graph. The introduction of path-level information enriches the model's modeling capability for complex associations between entities in the knowledge graph, aiding in more accurate judgment of triple confidence. Although the detailed investigation of path-level information is not the primary focus of this study, this approach does provide an effective means to enhance the accuracy of knowledge graph quality assessment.

Conclusions and future works

During the construction process of knowledge graphs, it is inevitable to introduce noise and conflicts. In response to the quality assessment problem of cybersecurity knowledge graphs, we manually built a cybersecurity knowledge graph dataset called CS13K. Subsequently, we extended the UCO ontology and created a new ontology specifically for the cybersecurity domain. Using this extended ontology, we leveraged Neo4j to construct the cybersecurity knowledge graph, providing support for cybersecurity situational awareness, threat identification, and prediction. Lastly, we proposed a Transformer-based model for knowledge graph

Table 10 The impact of embedding dimensions on the model's performance

Model	Dimension k				
	32	64	75	100	150
Tucker	0.671	0.837	0.860	0.894	0.893
AttTucker	0.720	0.877	0.899	0.908	0.908

Table 11 The evaluation results of each model

Dataset	Metrics	AttTucker	Path	AttTuc-Path
CS13K-N50	Accuracy	0.908	0.889	0.947

quality assessment, which demonstrated good evaluation performance under low-dimensional embeddings, showcasing the scalability of our model in large-scale knowledge graph datasets. Additionally, experimental results revealed that incorporating path-level information significantly enhanced the effectiveness of knowledge graph quality assessment. Specifically, on the CS13K-N50 dataset, our model achieved an accuracy of 0.947. In future, we will further expand our cybersecurity knowledge graph to encompass a broader range of cyber-attacks and threats. Moreover, we will explore methods for better integration of other types of data and information, such as cyber traffic data and intrusion detection system reports, to enhance the utility and comprehensiveness of our knowledge graph. We will also investigate more efficient knowledge representation algorithms to improve the performance of our knowledge graph quality assessment model.

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Data availability For the FB15K dataset used in the paper, it can be accessed through the following link: https://github.com/zqhead/TransE/tree/master/FB15k. Additionally, the CS13K dataset is available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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